Bernoulli NML Selects Relative-Periodic Domains

Residual defect masks expose persistent structure in cellular automaton spacetime

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Affiliation to be added

Main result. A single complexity-penalized selector fits a global periodic scaffold in 1D, 2D, and 3D cellular automata. The strongest 2D cases do not collapse to noise after background subtraction; they leave low-density residual masks that stay visibly structured.

Problem

Broad CA domains coexist with particles, interfaces, and other persistent defects. The practical task is to choose *which global period and shift best explain a finite observed spacetime.*

Poster lens. Fit the scaffold first, then study what survives in the residual.

Selector

- Translation by $(t, \mathbf{x}) \mapsto (t + p, \mathbf{x} + \mathbf{s})$ induces orbit classes.
- Majority vote on each orbit gives the nearest background $B^*(p, \mathbf{s})$.
- The residual mask is $M = U \oplus B^*(p, \mathbf{s})$.
- Bernoulli NML compares candidate periods and shifts without rewarding unnecessary template size.

$$\mathrm{NML}(p, \mathbf{s}) = \sum_j n_j H_b(\hat{\theta}_j) + \frac{1}{2} \sum_j \log_2 n_j$$

The fit term favors pure orbit classes. The penalty prevents larger templates from winning by sheer flexibility.

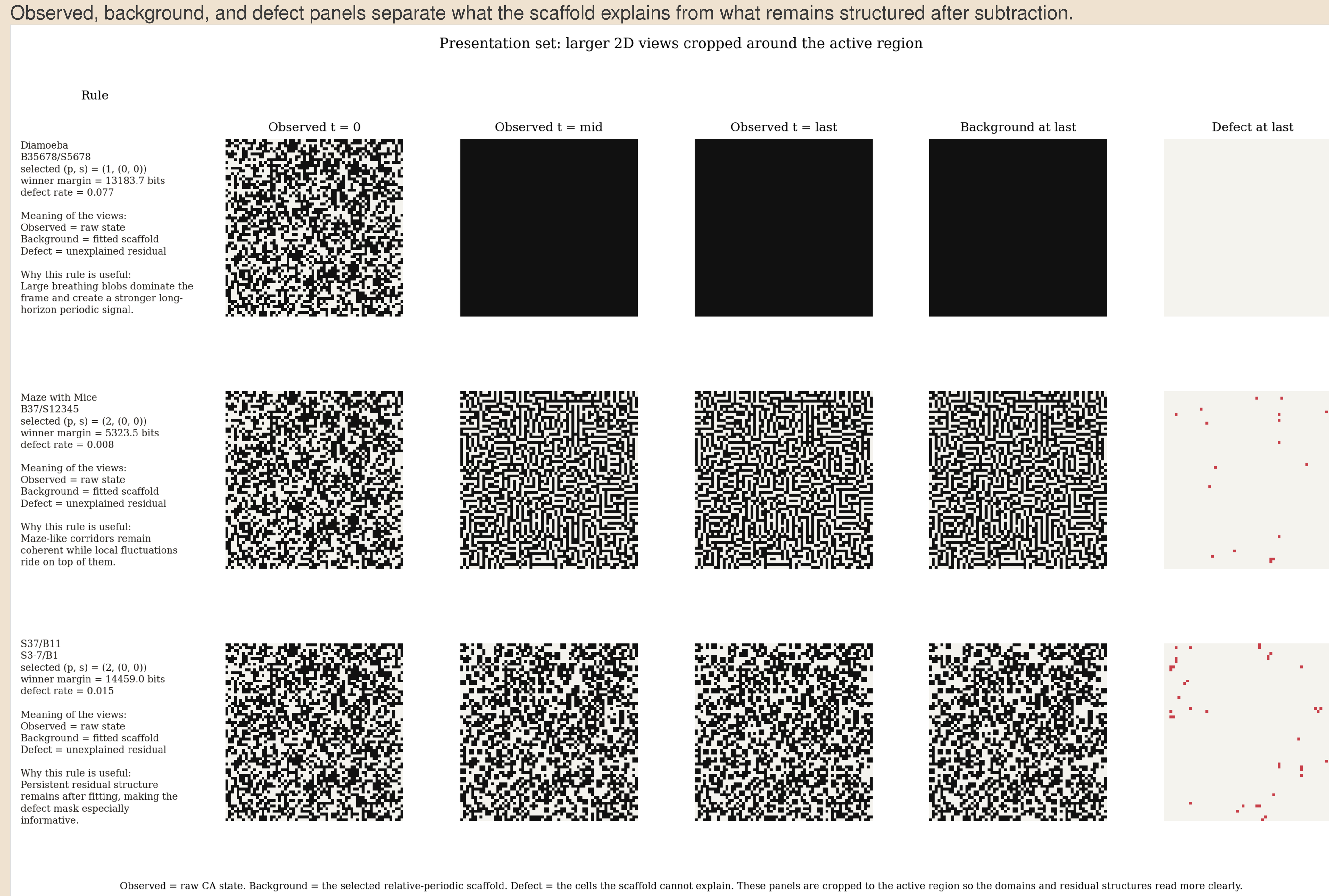
Why Not Raw Likelihood?

Selector	Outcome on 105 nontrivial LifeWiki rules at $T = 100$
Residual minimization	16 pick $p = 1$, 8 pick $p = 2$, and 81 pick period ≥ 4
Bernoulli NLL	All 105 pick the maximum tested period
Bernoulli NML	97 pick $p = 1$, 7 pick $p = 2$, and 1 picks $p = 8$

Bernoulli NML is the useful poster-side argument: without the complexity term, the selector overfits almost immediately.

2D Main Result

Observed, background, and defect panels separate what the scaffold explains from what remains structured after subtraction.



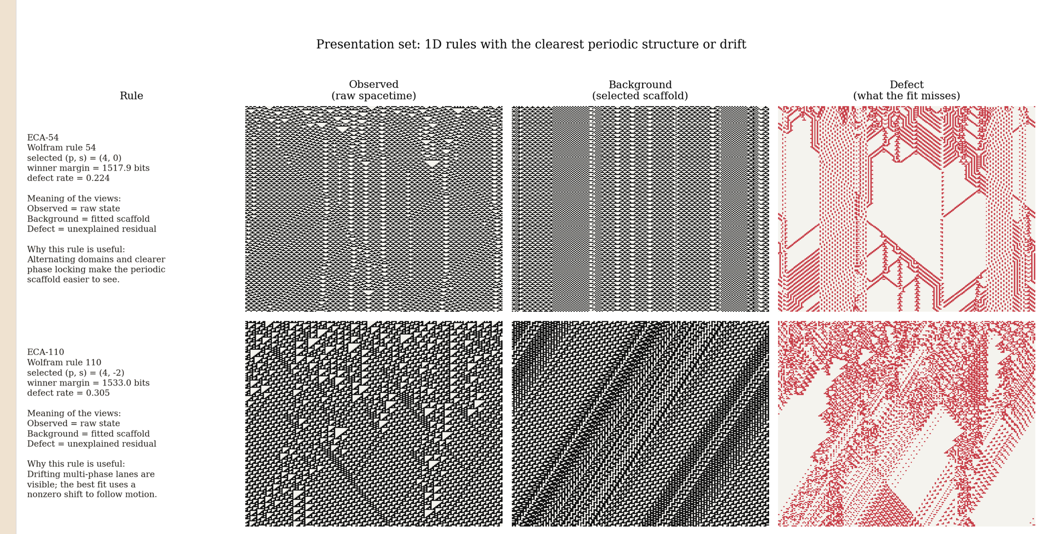
Observed = raw CA state. Background = the selected relative-periodic scaffold. Defect = the cells the scaffold cannot explain. These panels are cropped to the active region so the domains and residual structures read more clearly.

Most important visual claim. In 2D, the selected scaffold captures broad regular domains while residual masks preserve coherent, low-density structure. The method is therefore useful both for selection and for downstream defect analysis.

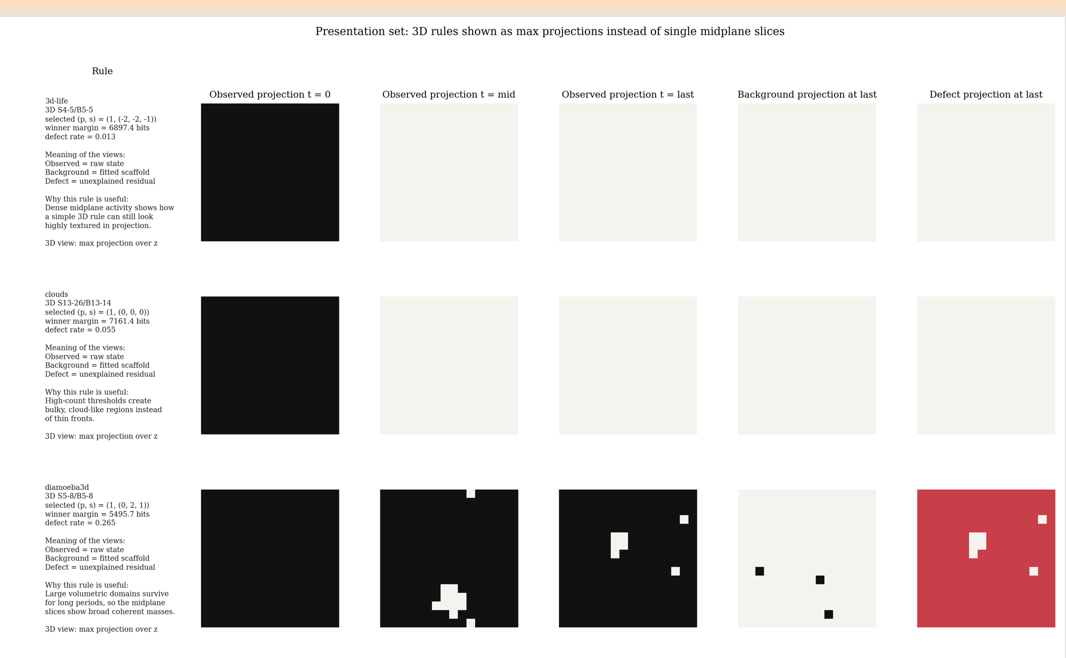
Focal 2D Cases

- **Diamoeba**: large breathing domains, selected winner $(p, \mathbf{s}) = (1, (0, 0))$, defect rate 0.077.
- **Maze with Mice**: highly coherent maze background, selected winner $(2, (0, 0))$, defect rate 0.008.
- **S37/B11**: low-defect background plus persistent residual structure, selected winner $(2, (0, 0))$, defect rate 0.015 in the presentation run.

Cross-Dimensional Checks



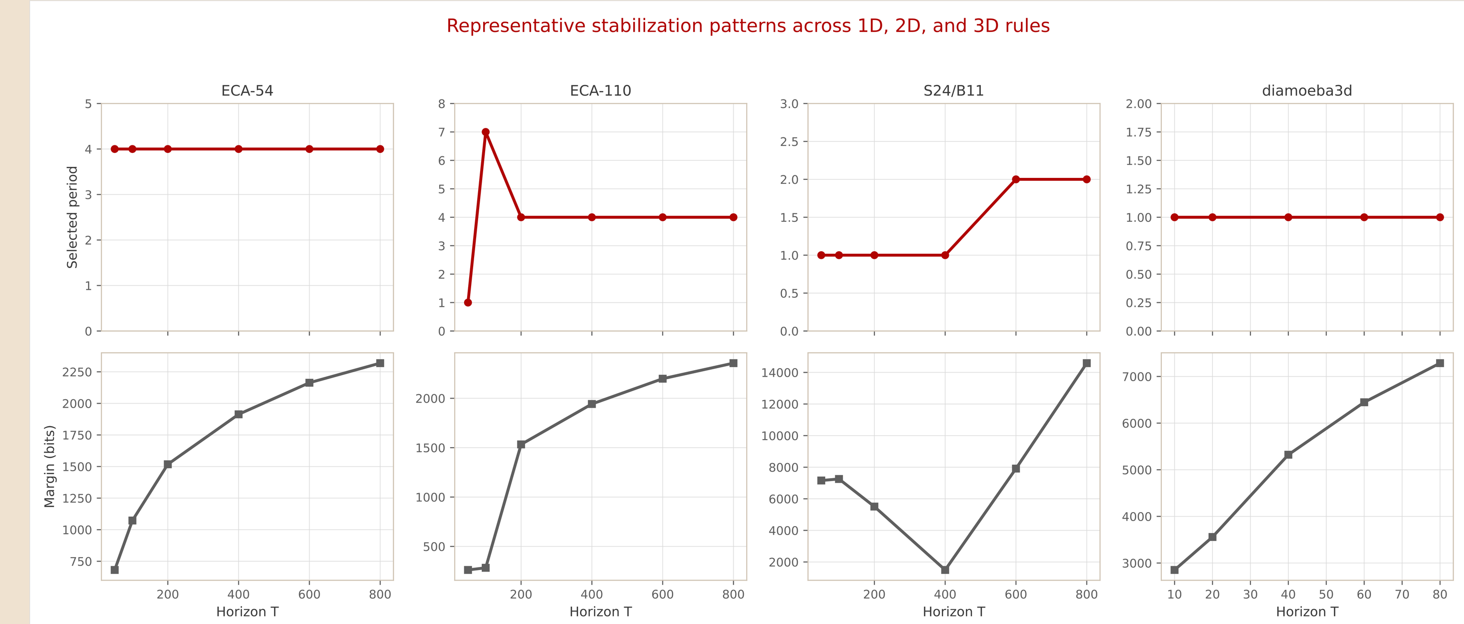
1D. ECA-54 locks to period 4. ECA-110 stabilizes to $(p, s) = (4, -2)$ under the period-first selector.



3D. The same decomposition still works on volumetric rules; the tested diamoeba3d case settles to period 1.

Stability Across Horizon

The selected period usually locks quickly as the observation horizon grows.



Top row: selected period versus horizon. Bottom row: winning margin in bits. The selected period either stabilizes immediately or changes only a small number of times before locking.

- **ECA-54** locks immediately to period 4.
- **ECA-110** passes through a short transient and then locks to period 4 with shift -2.
- **S24/B11** changes later, as longer horizons reveal additional periodic structure.
- **Diamoeba3d** corrects a small-sample transient and then locks to period 1.

Controls And Robustness

2D summary	Mean period	Mean defect rate
Original runs	2.2	0.014
Bernoulli i.i.d.	1.0	0.384
Space shuffled	1.0	0.384
Time shuffled	1.0	0.114

Shuffled and i.i.d. controls collapse back toward period 1 and much worse defect rates, which is consistent with the selector responding to real spacetime structure rather than generic occupancy statistics.

Rule	Modal p	Defect rate	Seed consistency
S14/B11	1	0.0059	10 / 10
S24/B11	2	0.0278	10 / 10
S37/B11	2	0.0441	10 / 10
S11/B37	4	0.0275	7 / 10

For **S37/B11**, the late defect density stays close to 0.011 across 32^2 to 192^2 grids, with mean late defects growing from 11.4 to 428.5.

Takeaways

- A single global selector gives usable periodic scaffolds in 1D, 2D, and 3D CA.
- The complexity penalty changes the selected period qualitatively; it is not cosmetic.
- In 2D, residual masks expose structured defects and make broad surveys feasible.
- The method is a front end for richer local analysis, not a replacement for computational mechanics.

Design notes for this version. The poster uses the ALIFE 2026 site palette: accent-1 beige background, accent-2 deep red, pale apricot highlight, and charcoal neutrals. Text was enlarged and sections were simplified to support fast scanning at distance.

Selected sources inside this repo. Algorithm figure: `poster/figures/algorithm_detailed_tikz.tex`. Stabilization figure: `scripts/alife_stabilization_summary.py`. Presentation panels: `outputs/alife_2026/rule_diagrams`. Control summaries: `outputs/alife_2026/null_controls`. Multi-seed and scaling summaries: `outputs/strengthening_v2`.

Project link. `https://github.com/zitterbewegung/relative_symmetry_repair`

Scan for repo, paper assets, and scripts.

Replace with a preprint URL if needed.